

The free energy of the spherical random perceptron

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Abstract

We prove an exact variational formula for the limiting pressure of the spherical random perceptron with an arbitrary bounded continuous single-pattern potential. The formula holds at every fixed positive density and inverse temperature, with convergence in expectation and in probability.

1 Introduction and statement

The spherical perceptron is a model of random constraints on a vector of fixed length. Its partition function measures the spherical volume of configurations weighted by their responses to independent random patterns. The question here is whether the pressure has an exact variational description for every fixed density and temperature, even when the single-pattern reward has no convexity, concavity, or symmetry.

Let $\alpha, \beta > 0$, let $\phi \in C_b(\mathbb{R}; \mathbb{R})$, and put $M_N = \lfloor \alpha N \rfloor$. For independent standard Gaussian vectors $g^1, \dots, g^{M_N} \in \mathbb{R}^N$, define

$$S_N = \{x \in \mathbb{R}^N : \|x\|^2 = N\}, \quad H_N(x) = \sum_{a=1}^{M_N} \phi\left(\frac{g^a \cdot x}{\sqrt{N}}\right), \quad p_N = \frac{1}{N} \log \int_{S_N} e^{\beta H_N(x)} d\sigma_N(x),$$

where σ_N is the uniform *probability* measure. The sign convention is positive log-partition pressure; the physical free energy per coordinate is $-p_N/\beta$.

Write $\mathcal{U}$ for the nondecreasing measurable functions $m : [0, 1] \rightarrow [0, 1]$ and set

$$D_m(t) = \int_t^1 m(s) ds, \quad S(m) = \frac{1}{2} \int_0^1 \left(\frac{1}{D_m(t)} - \frac{1}{1-t} \right) dt.$$

This is a nonnegative extended integral, with $1/0 = +\infty$. For $f = \beta\phi$, let

$$V_f(m) = \sup_v \mathbb{E}_B \left[f \left(B_1 + \int_0^1 m(t) v_t dt \right) - \frac{1}{2} \int_0^1 m(t) v_t^2 dt \right],$$

where B is standard Brownian motion with its usual filtration and the supremum is over progressively measurable real controls of finite expected displayed cost. This definition applies to bounded continuous f without terminal smoothness assumptions. In particular, $|V_f(m)| \leq \|f\|_\infty$.

Theorem 1. *For every $\alpha, \beta > 0$ and every $\phi \in C_b(\mathbb{R}; \mathbb{R})$,*

$$\lim_{N \rightarrow \infty} \mathbb{E} p_N = \inf_{m \in \mathcal{U}} \{\alpha V_{\beta\phi}(m) + S(m)\} =: \mathcal{P}(\alpha, \beta, \phi), \quad p_N \xrightarrow{\mathbb{P}} \mathcal{P}(\alpha, \beta, \phi).$$

Trials of infinite entropy have value $+\infty$. No condition that $m = 1$ near 1 is imposed.

The probability normalization of σ_N gives the elementary bounds

$$\alpha \mathbb{E} f(G) \leq \mathcal{P}(\alpha, \beta, \phi) \leq \alpha \log \mathbb{E} e^{f(G)}, \quad f = \beta \phi, \quad G \sim N(0, 1).$$

At finite N , Jensen's inequality inside the spherical integral gives $\mathbb{E} p_N \geq (M_N/N) \mathbb{E} f(G)$; applying it instead to the outer logarithm gives $\mathbb{E} p_N \leq (M_N/N) \log \mathbb{E} e^{f(G)}$. For a constant potential $\phi \equiv c$, both bounds coincide at $\mathcal{P} = \alpha \beta c$.

The problem and earlier formulas. Gardner's storage-volume problem introduced the statistical-mechanical study of the space of neural-network weights satisfying random constraints [7]. For nonnegative margins, Shcherbina and Tirozzi proved Gardner's replica-symmetric formula for a truncated logarithm of the feasible spherical volume below the storage threshold. They also treated a regularization with Gaussian smoothing and quadratic confinement in a restricted parameter regime [22, Theorems 2 and 3]. These hard-constraint and regularized observables differ from the arbitrary bounded reward in Theorem 1.

The finite-temperature functional itself has a substantial history. Györgyi and Reimann developed the continuous replica-symmetry-breaking prediction for a spherical perceptron: their equations (3.13), (3.17), and (3.19)–(3.22) give the terminal evolution and spherical entropy [10]. With physical potential $-\phi$ and positive log-partition pressure, these are the two terms in the formula above. Montanari and Zhou give a related finite-temperature replica derivation [11, Appendix B] in their study of exceptional projections of Gaussian point clouds. Their rigorous algorithmic theorem concerns approximate-message-passing values; the replica calculation in that appendix uses continuation in replica number and limit exchanges. The present theorem identifies the microscopic quenched pressure using ordinary Gibbs samples.

For an explicit normalization dictionary with [11], set $h = \phi$, $y = m$, and $\Psi = \beta f_y$ in their equation (100). Their functional A then satisfies $\alpha \beta A(m, \beta) = \alpha \Psi(0, 0) + S(m)$. Their dimension d and sample count n correspond to N and M_N : pressure per coordinate introduces the factor $n\beta/d \rightarrow \alpha\beta$.

Probabilistic methods and spherical entropy. The proof builds on the cascade, interpolation, and cavity methods developed for mean-field spin glasses. Ruelle's Poisson construction gives the hierarchical weights [21]; Guerra's interpolation supplies the classical replica-symmetry-breaking comparison [9]; and the extended variational principle of Aizenman, Sims, and Starr places fixed-size cavity increments at the center of the pressure calculation [1]. The spherical entropy is related to the Crisanti–Sommers functional [6] and its rigorous formulation by Talagrand [23, Equation (1.11)]. These Gaussian spin-glass results provide methods and comparison formulas; the nonlinear pattern Hamiltonian requires the arguments given below.

The obstacle and the main ideas. Although the patterns are Gaussian, their nonlinear sum $H_N(x)$ is generally not a Gaussian process in x . A Gaussian mixed-spin pressure formula therefore does not directly give Theorem 1. Nor does an arbitrary ϕ provide a convexity sign for a usual comparison interpolation. The proof uses two different kinds of increments: adding a random pattern, and adding spin coordinates.

The overlap $R(x, x') = x \cdot x'/N$ measures the agreement of two Gibbs samples. Small Gaussian perturbations impose the Ghirlanda–Guerra identities on its limiting array. The Dovbysh–Sudakov representation and Panchenko's support and ultrametricity theorems then describe that array by an ordered overlap law [15, 16, 18]. Finite Ruelle cascades evaluate functions of this law. We prove the sampling and marked transformations needed here, with credit to the joint marked-law calculus

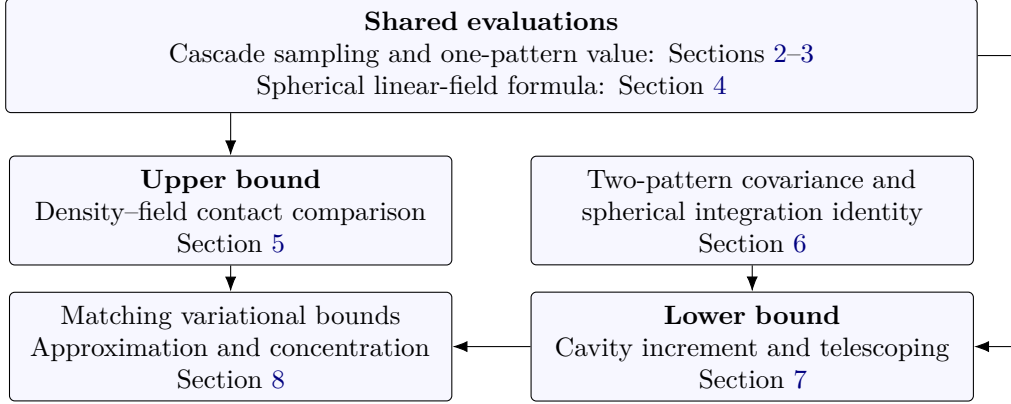


Figure 1: The two bounds share the cascade one-pattern value and the spherical linear-field calculation. The upper argument uses a contact comparison; the lower argument identifies the covariance generated by added coordinates. Arrows indicate the main proof inputs; they do not specify exchanges of limits.

of Panchenko and Talagrand [20]. The independent Gaussian remainder on each replica visit is retained even when two visits select the same leaf.

The upper bound adapts Mourrat’s Hamilton–Jacobi approach to nonconvex spin glasses. Enrichment by a cascade Gaussian field is used in the work of Mourrat and Panchenko [14, Equation (1.6) and Theorem 1.1]. The comparison here follows Mourrat’s contact-point supersolution argument [12, Section 4, proof of Theorem 4.1]; its vector-spin extension appears in [13, Theorem 3.4 and Section 5]. In this argument, perturbation parameters are selected at a contact minimum, where curvature and concentration enforce joint overlap identities. These identities order the spin and label overlaps. We use Panchenko’s synchronization argument [19, Section 4, Lemma 2] in the monotone-coupling formulation of Mourrat [12, Proposition 5.2 and Theorem 5.3]. The cited pressure bounds concern Gaussian Hamiltonians. For the nonlinear pattern Hamiltonian, we use Poissonized density as the time variable and the tail-order property of the one-pattern value to obtain the comparison.

For the lower bound, remove two patterns and restore them as independent marks. Their first and second moments show that the covariance of the field acting on new coordinates is a deterministic function of the spin overlap. Tangential integration on the sphere identifies that function with the derivative of spherical entropy. The ensuing cavity increment is therefore bounded below by the same variational functional. A uniform spherical-field calculation allows the cavity size to tend to infinity after the bulk limit. Identifying the cavity parameters through overlap observables is related to the critical-point approach of Chen and Mourrat [5, Sections 6–7]. Their results concern Gaussian vector-spin models and do not directly apply here; the two-pattern removal and spherical integration identity provide the relations needed in this model. Figure 1 shows where the two bounds share inputs.

Proof structure. Sections 2 and 3 develop the cascade calculus and an order property of the one-pattern value. Section 4 evaluates the spherical linear-field term, including the uniformity required in the cavity limit. Section 5 proves the upper bound by an affine comparison for negative pressure. For the lower bound, Section 6 uses two-pattern removal to identify bulk gradient covariances and a spherical integration identity to determine them explicitly. Section 7 evaluates and telescopes cavity increments. Section 8 passes from smooth potentials to bounded continuous ones and proves

concentration.

2 Quantiles and cascade preliminaries

The upper comparison and the cavity argument both evaluate the effect of a fresh Gaussian pattern on a limiting Gibbs measure. We first encode its overlap law by a quantile and a finite cascade. The marked-cascade calculation then specifies exactly what is shared by replicas and what must be sampled afresh at each visit.

Throughout the proof, $f = \beta\phi$. Sections 2–7 assume $f \in C_c^\infty(\mathbb{R})$; Section 8 removes this assumption. All perturbations and enrichments of the model in the proof will have their own auxiliary randomness, independent of the patterns unless stated otherwise. Expected log-partition functions for those models include expectation in this randomness (with parameter values held fixed, unless a further averaging in parameters is specified).

Up to changes irrelevant to the functionals, a trial m is the cumulative distribution function, restricted to $[0, 1)$, of a probability μ on $[0, 1]$ (put any remaining mass at 1). Let q be a nondecreasing quantile function, viewed modulo null sets on $(0, 1)$. We will use $V(q)$ and $S(q)$ for the value and entropy defined above in this parametrization; $D_q = D_m$. In particular

$$D_q(t) = 1 - t - \int_0^1 (q(u) - t)_+ du.$$

In scalar products of functions of the quantile parameter u , the integration is on $(0, 1)$ with Lebesgue measure.

Here is some notation for finite quantiles. Use $k \geq 1$ consecutive intervals of lengths $w_i > 0$, $0 \leq i \leq k-1$, of sum 1. Write $\zeta_i = \sum_{j < i} w_j$, including $\zeta_0 = 0$, $\zeta_k = 1$. A step q has values $0 \leq q_0 \leq \dots \leq q_{k-1} \leq 1$; set $q_{-1} = 0$, $q_k = 1$. Another kind of nonnegative nondecreasing function we will use, with no bound by 1 required, is h describing field parameters; similarly write h_i for step values on such intervals.

We use actual Gibbs samples (replicas), i.i.d. given the Gibbs measure, without any continuation in replica number; $\langle \rangle$ denotes Gibbs expectation, including for arbitrarily many such samples. The spin overlap on S_N is $R(x, x') = x \cdot x' / N$, giving an array $R_{\ell j}$. In enriched models replicas include extra labels.

2.1 Overlap identities and finite cascades

For the intervals above, use a Ruelle probability cascade (RPC) [21, Section 3] of depth $k-1$ with parameters $\zeta_1, \dots, \zeta_{k-1}$. Its leaf weights v_γ , summing to 1, are obtained by normalizing the products of Poisson weights along paths: for edges from each vertex at level $i-1$ to its children at level i use an independent Poisson process on $(0, \infty)$ of intensity $\zeta_i x^{-1-\zeta_i} dx$, whose points can be arranged in decreasing order. At depth zero there is just weight 1. For two leaves the common level is the depth $\gamma \wedge \eta$ of their last common vertex, including $k-1$ if equal.

We use the standard overlap-array theory developed in [17, 18]. The deep input is Panchenko's ultrametricity theorem [18, Theorem 1]. We state precisely the other facts needed and explain their use below. The Ghirlanda–Guerra (GG) law [8] for an array B with constant diagonal says that, given entries among the first n replicas, $n \geq 2$, the conditional law of $B_{1,n+1}$ is

$$\frac{1}{n} \mathcal{L}(B_{12}) + \frac{1}{n} \sum_{2 \leq j \leq n} \delta_{B_{1j}}.$$

This is an identity for the law that includes disorder averaging. We use Panchenko's ultrametricity theorem: a weakly exchangeable random Gram array (symmetric positive semidefinite in finite restrictions, bounded here, with fixed diagonal normalized or rescalable to 1) satisfying the GG identities has ultrametric off-diagonal overlaps, $B_{23} \geq \min(B_{12}, B_{13})$ a.s. For precision, the Dovbysh–Sudakov representation [15, Proposition 1] represents a normalized Gram array as $h^\ell \cdot h^j + a_\ell \mathbf{1}_{\{\ell=j\}}$, with $a_\ell \geq 0$ and a directing measure on the Hilbert unit ball. The support theorem [16, Theorem 2(a)], applied to the off-diagonal GG identities, makes $\|h^\ell\|^2$ equal to the supremum of the off-diagonal support almost surely. The represented inner-product array thus has deterministic diagonal; the off-diagonal identities give its full GG identities, to which the ultrametricity theorem applies. This verifies the Gram-array formulation without identifying the original diagonal with the directing-vector norm. Here one can take entries off the diagonal nonnegative for this theorem: nonnegativity itself follows under the GG law in these assumptions. Indeed if an event $B_{12} < -\delta$ with $\delta > 0$ has mass $b > 0$, the all-pairs off-diagonal event of this kind among n replicas can be extended to $n + 1$ with conditional probability at least b on this event, by exchangeability, the GG law and a union bound. This for arbitrarily large n contradicts positivity of the quadratic form on the all-ones vector.

We also use the standard finite RPC overlap result: for replicas sampled from its weights (and averaging the cascade randomness), their off-diagonal common-level array has two-replica level probabilities w_i and satisfies the GG identities. In particular a scalar array with values in $[0, 1]$ and diagonal 1 satisfying the Gram, exchangeability and GG assumptions above can be evaluated by finite cascades after rounding its off-diagonal overlaps by a nondecreasing finite-valued map into $[0, 1]$. The law of the rounded array off the diagonal is the one with overlaps given by values at the RPC levels, using the ascending list of values of positive probability and their probabilities. Indeed the rounded array is still ultrametric and satisfies GG. These properties determine its law with the given two-replica probabilities: given the first n , at each overlap threshold in the list, for each existing equivalence class at that threshold the chance the next replica belongs to it is specified by the GG formula using a representative. (Equivalence here means equal indices or overlap at least the threshold.) The nested classes and all these conditional probabilities determine the probabilities of all possible joining locations, hence of the new overlaps. This gives uniqueness starting with the two-replica law.

The finite-cascade sampling law. The complete weights-and-ancestry law was identified by Bolthausen and Sznitman [3, Theorem 2.2 and Proposition 1.4]. Here is a direct verification of the RPC law just used, including the conditioning on the whole sampled tree. Write $D = k - 1$ and $z_i = \zeta_i$, so $0 = z_0 < z_1 < \dots < z_D < z_{D+1} = 1$. At a vertex whose outgoing Poisson exponent is b , absorb each independent descendant total $T(C)$ into its edge weight: $y = uT(C)$. The marked intensity becomes

$$by^{-1-b} dy T(C)^b \mathbb{P}(dC).$$

The descendant subtrees therefore have independent laws biased by their totals to power b , independently of the new edge masses. An incoming bias by the parent total to power a , where $0 \leq a < b$, depends only on the sum of those masses and preserves this independence. The required moments are finite because the next Poisson exponent is strictly larger; at leaves the descendant total is one.

For stable Poisson jumps of exponent b , let S be their sum and bias their law by $S^a/\mathbb{E}S^a$. The probability of a specified partition of $n \geq 1$ visits into j nonempty blocks of sizes $n_1, \dots, n_j$ is

$$\frac{\Gamma(1-a)}{\Gamma(n-a)} \prod_{\ell=1}^{j-1} (\ell b - a) \prod_{r=1}^j \frac{\Gamma(n_r - b)}{\Gamma(1-b)}.$$

To obtain the formula, insert $S^{a-n} = \Gamma(n-a)^{-1} \int_0^\infty t^{n-a-1} e^{-tS} dt$ in the sum over distinct sampled Poisson points, use the Poisson factorial moment identity, and then integrate with the stable Laplace transform $\mathbb{E}e^{-tS} = e^{-ct^b}$. All terms are nonnegative, so these integrations are justified by monotone convergence. Ratios of the displayed probabilities give $(n_r - b)/(n - a)$ for joining an occupied child and $(jb - a)/(n - a)$ for taking a new child; the first visit is assigned to a new child.

Iterating the independent subtree decomposition gives these local split laws with $(a, b) = (z_{i-1}, z_i)$. The likelihood of a sampled nested partition factors over its occupied vertices. Conditional on the visits to a child, observations below that child concern its independent subtree; they therefore give no further information about the parent's normalized masses. Thus the same local ratios hold given the entire sampled partition. If $C_0 \supset C_1 \supset \dots \supset C_i = C$ is the path to an occupied depth- i class, with $n_{C_0} = n$, its next-visit probability is

$$\prod_{j=1}^i \frac{n_{C_j} - z_j}{n_{C_{j-1}} - z_{j-1}} = \frac{n_C - z_i}{n}.$$

With one existing replica this probability is $1 - z_i$; differences give the pair-level probabilities w_i . For a class containing replica 1, the displayed ratio is exactly the GG mixture for its threshold indicator. These indicators span all functions of the finite common level, proving GG with arbitrary tests of the previously sampled tree. The conditioning is on the nested partition of replica indices, not on numerical Poisson ranks or realized marks. Depth zero is deterministic; the endpoint exponents zero and one do not create Poisson branching levels.

2.2 Marked cascade transformations

We next describe what happens to both weights and marks when a cascade is reweighted. The joint law is needed when two removed patterns are restored: invariance of the weights alone would not identify products of their observables. The marked Poisson calculation is developed in [20, Lemmas 2.1 and 3.1, Corollary 3.2]; we give the conditional version needed here.

Marks may include common root data, independent of all Poisson weights. Given ancestor marks, children receive independent marks according to a probability kernel, also independently of the Poisson weights. For example, a Gaussian field with nonnegative nondecreasing covariance levels b_i has independent vertex increments of variances b_0 at the root and $b_i - b_{i-1}$ at depth $i \geq 1$. Let X_{k-1} be a leaf log factor depending on the marked path and define backward, at each vertex,

$$X_{i-1} = \zeta_i^{-1} \log E_i e^{\zeta_i X_i}, \quad i = k-1, \dots, 1, \quad (1)$$

where E_i integrates the child mark given the path through depth $i-1$. We assume these recursions are finite, with integrable root value whenever an unconditional expectation is taken.

Lemma 2 (Marked cascade law). *In the preceding setting, put $L_X = \log \sum_\gamma v_\gamma e^{X_{k-1}(\gamma)}$. The following conclusions hold.*

1. *Conditional on root data,*

$$E[L_X \mid \text{root}] = X_0, \quad E[(L_X - X_0)^2 \mid \text{root}] \leq C_\zeta,$$

where C_ζ depends only on the fixed cascade parameters.

2. *After multiplication by $e^{X_{k-1}}$ and normalization, the weights, up to tree relabeling, again have the ordinary cascade law. Jointly with these weights, the child mark kernels are tilted by $e^{\zeta_i(X_i - X_{i-1})}$. The transformed unmarked tree is independent of root data. Given a finite sampled tree shape, the root has its original law and the marks are generated by these tilted kernels, independently for distinct children conditional on their ancestors.*

3. Suppose the marks consist of independent components: their root variables are independent, their child kernels factor over components with each kernel depending only on its own ancestral component marks, and X_{k-1} is a sum of functions of the separate component paths. Then the recursions are additive and the tilted kernels factor. In particular the components remain independent conditional on a sampled finite tree shape.

Proof. Multiply each level- i edge weight by $e^{X_i - X_{i-1}}$ and reorder the children. The conditional fractional moment

$$E_i e^{\zeta_i(X_i - X_{i-1})} = 1$$

shows that Poisson mapping preserves the edge intensity and tilts its mark kernel by precisely this factor. Only this fractional moment is required; see also [3, Appendix, Proposition A.2]. A first moment of $e^{X_{k-1}}$ is unnecessary, a distinction that will matter for the canonical quadratic factors in Section 4.

The mapping is applied from the leaves upward. Its induction preserves the ordinary law of the entire transformed unmarked subtree, independently of ancestor marks. Indeed, at a parent regard the child mark together with its already transformed unmarked descendant subtree as one mark. By induction the latter has a fixed law independent of the path. The parent multiplier depends only on the path and child mark, so mapping gives the ordinary edge intensity times the tilted child kernel times the unchanged descendant law. The unit fractional moment removes any scale depending on ancestors. This proves the joint-law assertion. Sampling labels depends only on the unmarked tree; conditioning on their shape therefore leaves the root law and the stated mark kernels intact. For independent components, the exponential moments in (1) factor at each step, proving the last assertion. Independence of the root components is part of this hypothesis; conditional independence given shared random root data alone would not suffice after averaging those data.

For the first assertion, let T and T' be the original and transformed sums of unnormalized leaf products. The product of edge multipliers along a path is $e^{X_{k-1} - X_0}$, hence

$$L_X = X_0 + \log T' - \log T.$$

Conditional on the root, both totals have the ordinary cascade total's law. That total is positive and finite and has finite log second moment. To see this, sum descendant totals one level at a time: a Poisson sum of exponent ζ_i with independent positive factors of finite ζ_i -moment is positive stable of index ζ_i , up to scale. Successive exponents increase, so the necessary moments exist. A positive stable variable has positive moments of every smaller order and finite negative moments, the latter also following from its Laplace transform. These imply the claimed log moment. The two conditional marginal laws therefore give zero mean and a fixed second-moment bound for $\log T' - \log T$; independence of the totals is not needed. At depth zero $L_X = X_0$ and no Poisson calculation is required. $\square$

There is one further distinction when Gaussian marks represent replica arrays. Suppose the shared field $Z(\gamma)$ has covariance levels b_i and the desired diagonal variance is $d \geq b_{k-1}$. Its evaluations on replica visits are

$$G^\ell = Z(\gamma^\ell) + \sqrt{d - b_{k-1}} \eta_\ell, \quad (\eta_\ell)_{\ell \geq 1} \text{ independent standard Gaussians.} \quad (2)$$

The residuals are independent of the shared field and labels, and remain independent even when two visits choose the same leaf. Thus distinct replicas then have covariance b_{k-1} , while each has variance d . For a leaf factor, integrate the residual before applying (1). After reweighting, an observable on a visit uses the ordinary tilt of that residual's law, separately for each visit. More generally the same rule applies to any fresh terminal marks averaged inside a leaf factor.

2.3 Passing fresh fields through overlap limits

The following passage explains why finite cascades evaluate the fresh patterns and cavity fields of a limiting overlap array. Here the base Gibbs measure may depend on disorder, and a fresh Gaussian field may have a covariance determined by that base disorder.

Lemma 3 (Bounded fresh-field passage). *Suppose that the covariance arrays of finitely many conditionally centered Gaussian fresh fields, together with the needed bounded base observables, are uniformly bounded and converge in law on every finite set of base replicas. Conditional on base data and replicas, the fields have the specified joint Gaussian law; different components are conditionally independent when so specified. Let W be a bounded continuous function of the fresh marks and base observables of one replica, with fixed bounds $0 < a \leq W \leq b < \infty$. Then both the expected log integral $E \log \langle W \rangle$ and bounded continuous tests of the fresh evaluations and converging base observables under reweighting by W pass to limits determined by those finite array laws. The same holds after adjoining an independently integrated trial spin with a fixed compactly supported law.*

Proof. For a bounded continuous test Ψ of n replicas, the reweighted expectation is

$$E \left[\frac{\langle \Psi \prod_{\ell=1}^n W(x^\ell) \rangle}{\langle W \rangle^n} \right]. \quad (3)$$

Approximate s^{-n} and $\log s$ uniformly by polynomials on $[a, b]$. Each resulting term is an expectation on finitely many base replicas, with the additional replicas evaluating the same fresh fields. Such terms converge because finite-dimensional Gaussian expectations are continuous in the covariance matrix, including degenerate matrices. The uniform approximation error then proves both assertions. Extra trial-spin integration can be included in the reference probability measure and in the same calculation. $\square$

For the pattern and cavity fields below, the limiting off-diagonal covariances are continuous nonnegative nondecreasing functions of a scalar ultrametric GG overlap, bounded by their respective diagonal variances. Round that overlap down by nondecreasing finite-valued maps with error tending uniformly to zero, retaining the true field diagonals. The rounded overlap law and these ordered covariance levels have the finite-cascade representation above. If a fresh field's diagonal exceeds its top off-diagonal covariance, use (2) in each polynomial term. Its independent residuals are thereby averaged at the leaf before the marked recursion. This is forced by the finite-replica covariance matrix, rather than a choice of a shared leaf mark. As the rounded covariances approach their original values, Gaussian continuity and the uniform polynomial error give the desired limit. Section 7 supplies a separate truncation estimate for its unbounded spherical factor.

2.4 Gaussian differentiation at fixed size

We will also differentiate partition integrals at fixed dimension, before taking overlap limits. Let ρ be a reference probability measure, possibly random, independent of the added Gaussian fields conditional on its base data. For centered Gaussian energy G_s whose bounded covariance K_s varies affinely with s , Gaussian integration by parts gives

$$\frac{d}{ds} E \log \int e^{G_s(x)} d\rho(x) = \frac{1}{2} E \left\langle \dot{K}_s(x^1, x^1) - \dot{K}_s(x^1, x^2) \right\rangle_s. \quad (4)$$

Explicit deterministic shifts are differentiated separately. This formula, its integral along the covariance segment, and its one-sided endpoint versions apply to the bounded-spin, finite-cascade

integrals used below. Ordinary coupling derivatives, including Gaussian insertions and products of Gibbs averages with bounded continuous overlap tests, may likewise be passed through expectation.

Here are moment bounds justifying these uses despite the infinitely many cascade leaves. For $r \geq 1$, Jensen bounds positive powers of a Gaussian exponential integral by the integral of e^{rG} . For any $r > 0$ it also gives

$$\left(\int e^G d\rho \right)^{-r} \leq \exp \left(-r \int G d\rho \right) \leq \int e^{-rG} d\rho.$$

Uniform covariance bounds and Gaussian exponential moments therefore control positive and inverse moments, also with polynomial Gaussian insertions by Hölder's inequality. Bounded pattern energies independent of the added fields may be absorbed into ρ after normalization; their normalization has the needed moments also for a Poisson number of patterns. These estimates hold locally uniformly in bounded coupling parameters at each fixed dimension.

To justify (4) directly, first replace ρ by its empirical measure from independent samples conditional on ρ , also independent of the fresh Gaussian fields. For the spheres and countable leaf coordinates in this paper, the state space is separable and the fields are continuous in the spin coordinate at each leaf. Conditional laws of large numbers give convergence of the weighted integrals and their continuous replica tests; polynomial insertions can first be truncated. The moment bounds pass these statements through expectation. Couple the fields on a segment as $\sqrt{1-s}G_0 + \sqrt{s}G_1$. Exponentials involving $|G_0| + |G_1|$, with polynomial factors when needed, give uniform moment bounds, and the bounded covariance terms on the right side of (4) are continuous in s . The integral identity thus passes from empirical measures by bounded convergence in s , including one-sided derivatives at the endpoints. The same local bounds give ordinary coupling differentiation and uniform integrability. The canonical integral with unbounded spins in Section 4 is instead treated by its explicit recursion.

3 The single-pattern value and its order property

Adding one independent pattern changes the log partition function by an expected log Gibbs integral. We first identify that integral with the control value $V(q)$, then prove the tail-integral order that supplies the sign in the upper comparison. Throughout this section $f \in C_c^\infty(\mathbb{R})$, as stipulated in Section 2.

The cascade integral and control representation. Fix a finite quantile with the notation q_i, w_i, ζ_i of Section 2. On its cascade let $z(\gamma)$ be a centered Gaussian mark with covariance $q_{\gamma \wedge \eta}$, independent of the cascade weights. We claim that

$$V(q) = \mathbb{E} \log \sum_{\gamma} v_{\gamma} \mathbb{E}_G \exp f(z(\gamma) + \sqrt{1 - q_{k-1}} G), \quad G \sim N(0, 1).$$

Here the outer expectation includes the cascade and all its shared Gaussian marks. The inner expectation averages the residual noise of one replica visit. Different visits use independent copies of G , even if their labels coincide. Thus the shared field has covariance $q_{\gamma \wedge \eta}$ between distinct replicas, while every replica's total field has variance one.

For $s \geq 0$ and $d \geq 0$, write

$$\mathcal{T}_{s,d}g(x) = \begin{cases} d^{-1} \log \mathbb{E} \exp(dg(x + \sqrt{s}G)), & d > 0, \\ \mathbb{E}g(x + \sqrt{s}G), & d = 0. \end{cases}$$

The operator $\mathcal{T}_{0,d}$ is the identity, so zero-length intervals require no separate convention. Starting with $U_k = f$, define backward functions by

$$U_{i-1} = \mathcal{T}_{q_i - q_{i-1}, \zeta_i} U_i, \quad i = k, k-1, \dots, 0.$$

Recall that $q_{-1} = 0$, $q_k = 1$, $\zeta_0 = 0$, and $\zeta_k = 1$. The first backward step, from 1 to q_{k-1} , is the logarithm of the residual average in the displayed cascade integral. The marked-cascade recursion handles levels $k-1$ through 1, and the final ordinary expectation averages the root mark of variance q_0 . Consequently that integral equals $U_{-1}(0)$.

To identify this number with the control value, interpolate the recursion on each interval $[q_{i-1}, q_i]$ by

$$U(t, x) = \mathcal{T}_{q_i - t, \zeta_i} U_i(x).$$

These pieces agree at their endpoints and solve the backward equation

$$\partial_t U + \frac{1}{2} \partial_{xx} U + \frac{m(t)}{2} (\partial_x U)^2 = 0, \quad U(1, x) = f(x),$$

where $m(t) = \zeta_i$ in the interior of that interval. The heat semigroup and exponential transform give this equation directly. Differentiating the Gaussian formulas shows that the first and second spatial derivatives are bounded, including their limits at interval endpoints, for the fixed finite recursion.

For an admissible control v , set

$$X_t^v = B_t + \int_0^t m(s) v_s ds.$$

Itô's formula on successive intervals and completion of the square give

$$\mathbb{E} \left[f(X_1^v) - \frac{1}{2} \int_0^1 m(t) v_t^2 dt \right] = U_{-1}(0) - \frac{1}{2} \mathbb{E} \int_0^1 m(t) (v_t - \partial_x U(t, X_t^v))^2 dt.$$

Bounded derivatives and the finite expected control cost justify the expectations, or one may first localize. This proves the upper bound by $U_{-1}(0)$ for every control. Equality is attained with $v_t = \partial_x U(t, X_t^v)$: this feedback is bounded and Lipschitz in space, so its stochastic equation has an adapted solution of finite cost. Hence $V(q) = U_{-1}(0)$, proving the claimed cascade identity. Brownian variational formulas [4, Theorem 3.1] and the control formulation of the Parisi PDE [2, Theorem 3] provide the background for this representation; the calculation above verifies it for the arbitrary smooth terminal function needed here.

Continuity and general overlap laws. For a Lipschitz terminal function with $L = \text{Lip}(f)$, the control supremum can be restricted to $|v| \leq L$. Indeed clipping v to this interval changes the terminal reward by at most $L \int m|v - \text{clip}(v)|$, while the saved quadratic cost is at least this quantity. Comparing the same clipped controls for m and m' therefore gives

$$|V_f(m) - V_f(m')| \leq \frac{3}{2} L^2 \|m - m'\|_1.$$

Area integration between distribution functions and their quantiles gives $\|m - m'\|_1 = \|q - q'\|_1$, so V is also continuous in the L^1 distance between quantiles.

Now suppose a sequence of base Gibbs measures has a limiting overlap array satisfying the scalar Gram, exchangeability, and GG assumptions, with diagonal one and off-diagonal quantile q . A fresh pattern has conditional covariance equal to that overlap array. Its factor e^f is continuous, bounded

above, and bounded away from zero, so Lemma 3 passes its expected log Gibbs integral to the overlap limit. Round the off-diagonal overlaps by finite-valued monotone maps whose uniform error tends to zero, and keep the diagonal equal to one. The resulting cascade integrals equal V of the rounded quantiles by the preceding calculation, with residual noise on every visit. Their limits equal $V(q)$ by L^1 continuity. This identifies the one-pattern increment for general limiting overlap laws.

Tail-integral order. For quantiles p, q taking values in $[0, 1]$, we will prove

$$\int_s^1 p(u) du \geq \int_s^1 q(u) du \quad (0 \leq s \leq 1) \implies V(p) \leq V(q). \quad (5)$$

First take finite steps on the same intervals. We use the tilted Gaussian chain associated with the backward recursion. Start at $Z_{-1} = 0$. Given $Z_{i-1} = x$, the transition to Z_i has the Gaussian law $N(x, q_i - q_{i-1})$ multiplied by

$$\exp\{\zeta_i(U_i(Z_i) - U_{i-1}(x))\}, \quad i = 0, \dots, k.$$

The recursion makes this density integrate to one; at $i = 0$ there is no tilt. This discrete chain records the successive marks under the cascade transformation. Differentiating the recursion in its spatial argument gives

$$U'_{i-1}(Z_{i-1}) = \mathbb{E}[U'_i(Z_i) \mid Z_{-1}, \dots, Z_{i-1}].$$

Thus $U'_i(Z_i) = \mathbb{E}[f'(Z_k) \mid Z_{-1}, \dots, Z_i]$ is a bounded martingale, and

$$c_i := \mathbb{E}[U'_i(Z_i)^2], \quad 0 \leq i \leq k-1, \quad 0 \leq c_0 \leq \dots \leq c_{k-1}.$$

Keeping the interval weights fixed and changing the times q_i by $\dot{q}_i$ has first variation

$$\dot{V}(q) = -\frac{1}{2} \sum_{i=0}^{k-1} w_i \dot{q}_i c_i.$$

Indeed lengthening the interval before q_i inserts $\frac{1}{2}(U''_i + \zeta_i(U'_i)^2)$, while shortening the interval after it subtracts $\frac{1}{2}(U''_i + \zeta_{i+1}(U'_i)^2)$. The second derivatives cancel and $\zeta_{i+1} - \zeta_i = w_i$. Differentiating the preceding operators propagates this difference by exactly the tilted expectations above. The Gaussian formulas and bounded spatial derivatives justify these differentiations. At coinciding times, one uses one-sided variations or moves permanently coinciding times together along an order-preserving line segment.

For the segment from q to p , put $a_i = w_i(p_i - q_i)$ and $A_i = \sum_{j \geq i} a_j$. The tail hypothesis at interval boundaries gives $A_i \geq 0$. At every point where the segment derivative is computed, its coefficients c_i are nonnegative and nondecreasing, so summation by parts yields

$$\sum_i c_i a_i = c_0 A_0 + \sum_{i=1}^{k-1} (c_i - c_{i-1}) A_i \geq 0.$$

The first variation is therefore nonpositive along the segment. Continuity at its endpoints proves (5) for finite steps. For general p, q , replace them by their averages on a common refining interval grid. These averages remain nondecreasing; their tail differences agree with the original ones at grid boundaries and interpolate linearly between boundaries, so the tail inequality is preserved. They converge in L^1 , and the already established continuity of V proves (5) in full generality.

Finite-step approximation. In minimizing the variational functional for smooth f , it suffices to use finite step quantiles with maximum strictly below one. Round q down on dyadic grids and truncate at $1 - 2^{-j}$. The resulting quantiles $q^{(j)}$ increase to q up to null sets. The values $V(q^{(j)})$ converge to $V(q)$ by L^1 continuity. The formula for D_q gives $D_{q^{(j)}} \downarrow D_q$, so monotone convergence in the nonnegative entropy integral gives $S(q^{(j)}) \uparrow S(q)$, including when the latter is infinite.

4 The spherical linear-field formula

The entropy term must agree with an actual spherical integral. We compute its dual linear-field pressure, first for a fixed finite cascade and then uniformly over overlap quantiles bounded above by a fixed $B < 1$. The latter estimate is needed when the cavity size tends to infinity. For a trial equal to one above $\hat{q} < 1$, our entropy is $S(m) = \frac{1}{2}[\int_0^{\hat{q}} D_m(t)^{-1} dt + \log(1 - \hat{q})]$, which agrees with the spherical entropy in [23, Equation (1.11)]. We derive the linear-field identity with the uniformity and endpoint conventions used here.

For a nonnegative nondecreasing step h with intervals as above, define

$$l_n(h) = \frac{1}{n} E \log \sum_{\gamma} v_{\gamma} \int_{S_n} \exp(z(\gamma) \cdot x - nh_{k-1}) d\sigma_n(x),$$

where z has n independent Gaussian cascade field coordinates with covariance levels $2h_i$. This functional is consistent under subdividing the parameter intervals and leaving the path h unchanged: this follows from the recursion, since at repeated field levels the new Gaussian increment is zero. Further

$$|l_n(h) - l_n(\tilde{h})| \leq \int_0^1 |h - \tilde{h}|.$$

Indeed use Gaussian covariance interpolation on common intervals, justified in Section 2.4. The direction derivative along the segment, with direction $d = \tilde{h} - h$, is $-E\langle R_{12} d_{\gamma^1 \wedge \gamma^2} \rangle$, with overlap on S_n . The label probabilities are w_i by the cascade transformation (log factors here include the log spherical integral). We can therefore extend l_n by continuity to bounded nonnegative nondecreasing h .

Proposition 4 (Spherical linear-field formula). *For fixed interval weights, uniformly on compact sets of nonnegative nondecreasing step field values,*

$$l_n(h) \longrightarrow \min_{q_0 \leq \dots \leq q_{k-1}; 0 \leq q_i < 1} \left\{ S(q) - \sum_i w_i h_i q_i \right\} \quad (6)$$

as $n \rightarrow \infty$. Furthermore, for each fixed $0 \leq B < 1$, if $0 \leq q(u) \leq B$ a.e. is nondecreasing and

$$2h(u) = A_q(q(u)), \quad A_q(r) := \int_0^r \frac{dt}{D_q(t)^2} \quad (0 \leq r < 1), \quad (7)$$

then

$$l_n(h) = S(q) - \int_0^1 h q + o(1), \quad (8)$$

where the error tends to zero uniformly over these q .

Proof. Entropy minimization. For a step q with maximum < 1 , put $D_i = D_q(q_i)$. Then

$$D_{i-1} - D_i = \zeta_i(q_i - q_{i-1}) \quad (1 \leq i < k), \quad D_{k-1} = 1 - q_{k-1}.$$

Differentiating $D_q(t) = 1 - t - \sum_i w_i(q_i - t)_+$ gives

$$\partial_{q_i} S(q) = \frac{w_i}{2} A_q(q_i).$$

The same coefficients give directional derivatives along admissible directions at tied or zero coordinates. The entropy integrand vanishes near 1 in a neighborhood of any such q , so the differentiation has no endpoint term.

Extend the objective in (6) to the closed ordered cube $0 \leq q_0 \leq \dots \leq q_{k-1} \leq 1$. It is lower semicontinuous by the nonnegative integral defining S and continuity of $D_q(t)$ in the positions. If $q_{k-1} = 1$, the overlap law has positive mass a at 1, whence $D_q(t) \leq (1-a)(1-t)$ and $S(q) = +\infty$. Since $q = 0$ has finite value, a minimum is attained with $q_{k-1} < 1$.

At a minimizer, consider the direction $d_i = h_i - A_q(q_i)/2$. It is admissible for sufficiently short positive steps: at a tie the $A_q(q_i)$ agree and the h_i are ordered, while at $q_i = 0$ one has $d_i = h_i \geq 0$. The derivative of the objective in this direction is $-\sum_i w_i d_i^2$. It must be nonnegative, so every $d_i = 0$. Thus the minimizer satisfies (7).

Canonical integral. We now evaluate $l_n(h)$ for *any* finite pair q, h satisfying (7) with $q_{k-1} < 1$. This will apply both to the minimizer just found and to the stationary fields in the second assertion. Set $b_* = 1/D_{k-1}$ and define the random canonical partition function

$$K_n(b) = \sum_{\gamma} v_{\gamma} \int_{\mathbb{R}^n} \exp\left(z(\gamma) \cdot x - \frac{b}{2} \|x\|^2\right) \frac{dx}{(2\pi)^{n/2}}.$$

The deterministic shift $-nh_{k-1}$ is omitted here. Write

$$b_{k-1} = b, \quad b_{i-1} = b_i - 2\zeta_i(h_i - h_{i-1}) \quad (1 \leq i < k).$$

Stationarity gives

$$2(h_i - h_{i-1}) = \frac{q_i - q_{i-1}}{D_i D_{i-1}}, \quad 2h_0 = \frac{q_0}{D_0^2}.$$

Consequently $b_i = 1/D_i$ at $b = b_*$, and all these precisions remain positive in a neighborhood of b_* .

Here the cascade recursion can be computed explicitly. Write z_i for the cumulative Gaussian field through level i . At the leaf, integration over x gives $X_{k-1} = -\frac{n}{2} \log b + \|z_{k-1}\|^2/(2b)$. The backward recursion preserves the form

$$X_i = c_i + \frac{\|z_i\|^2}{2b_i}, \quad c_{i-1} = c_i + \frac{n}{2\zeta_i} \log \frac{b_i}{b_{i-1}}, \quad c_{k-1} = -\frac{n}{2} \log b.$$

Indeed the level- i increment is an independent Gaussian vector of covariance $2(h_i - h_{i-1})I_n$. The conditional moment $E_i e^{\zeta_i X_i}$ in the recursion is finite exactly when $b_{i-1} > 0$. Thus these are the fractional-moment hypotheses of the marked-cascade transformation; no first moment of $e^{X_{k-1}}$ is needed. The root vector has independent coordinates of variance $2h_0$, so

$$\frac{1}{n} E \log K_n(b) = C(b) := \frac{1}{2} \left(-\log b + \sum_{1 \leq i < k} \frac{1}{\zeta_i} \log \frac{b_i}{b_{i-1}} + \frac{2h_0}{b_0} \right).$$

Moreover $n^{-1} \log K_n(b) \rightarrow C(b)$ in probability at every such b . The root value $c_0 + \|z_0\|^2/(2b_0)$ concentrates after division by n , and the remaining difference of log cascade totals has bounded conditional second moment for this fixed cascade, by Section 2. Finally,

$$-2C'(b_*) = D_{k-1} + \sum_{i=1}^{k-1} \frac{D_{i-1} - D_i}{\zeta_i} + q_0 = 1.$$

This derivative will force the canonical squared radius, divided by n , to concentrate at 1.

Spherical comparison. Let $Z_n^{\text{sph}}(v)$ be the weighted spherical integral without deterministic shift at squared radius nv , with probability surface measure and the same fields and weights. It is at least one and nondecreasing in $v \geq 0$, by symmetry and the cosh representation for each linear-field integral. Radial integration gives

$$K_n(b) = \int_0^\infty Z_n^{\text{sph}}(v) \rho_{n,b}(dv), \quad \rho_{n,b}(dv) = \frac{(n/2)^{n/2}}{\Gamma(n/2)} v^{n/2-1} e^{-bnv/2} dv.$$

This radial measure is not normalized. For a fixed small $\epsilon > 0$, Stirling's formula gives, on either band $I = [1, 1 + \epsilon]$ or $I = [1 - \epsilon, 1 + \epsilon]$,

$$\frac{1}{n} \log \rho_{n,b}(I) = \frac{1-b}{2} + O(\epsilon) + o_n(1).$$

Since $K_n(b_*) \geq Z_n^{\text{sph}}(1) \rho_{n,b_*}([1, 1 + \epsilon])$, taking expected logarithms, then $n \rightarrow \infty$ and $\epsilon \downarrow 0$, proves

$$\limsup_n l_n(h) \leq C(b_*) + \frac{b_* - 1}{2} - h_{k-1}.$$

For the reverse bound, normalize $Z_n^{\text{sph}}(v) \rho_{n,b_*}(dv)$ by $K_n(b_*)$ to obtain a random probability measure λ_n on the squared-radius variable v . For small $d > 0$ the two tails satisfy

$$\begin{aligned} \lambda_n([1 + \epsilon, \infty)) &\leq e^{-nd(1+\epsilon)/2} \frac{K_n(b_* - d)}{K_n(b_*)}, \\ \lambda_n([0, 1 - \epsilon]) &\leq e^{nd(1-\epsilon)/2} \frac{K_n(b_* + d)}{K_n(b_*)}. \end{aligned}$$

The normalized logarithms of the right sides converge in probability to, respectively,

$$C(b_* - d) - C(b_*) - \frac{d(1 + \epsilon)}{2}, \quad C(b_* + d) - C(b_*) + \frac{d(1 - \epsilon)}{2}.$$

Both equal $-d\epsilon/2 + o(d)$ because $C'(b_*) = -1/2$. Choose d small after fixing ϵ . It follows that $\lambda_n([1 - \epsilon, 1 + \epsilon]) \rightarrow 1$ in probability. Hence, with probability tending to one,

$$\frac{1}{2} K_n(b_*) \leq Z_n^{\text{sph}}(1 + \epsilon) \rho_{n,b_*}([1 - \epsilon, 1 + \epsilon]).$$

This gives a lower bound in probability for $n^{-1} \log Z_n^{\text{sph}}(1 + \epsilon)$ by $C(b_*) + (b_* - 1)/2 - O(\epsilon)$. Its nonnegativity transfers that bound to expectations: every smaller positive bound is exceeded with probability tending to one, and a nonpositive bound is automatic. Replacing squared radius n by $n(1 + \epsilon)$ rescales the field covariances by the same factor, so

$$\liminf_n (l_n((1 + \epsilon)h) + (1 + \epsilon)h_{k-1}) \geq C(b_*) + \frac{b_* - 1}{2} - O(\epsilon).$$

Letting $\epsilon \downarrow 0$ and using the common Lipschitz bound for l_n identifies its limit as $C(b_*) + (b_* - 1)/2 - h_{k-1}$.

Identifying the entropy. Integrating over the intervals between the positions q_i gives

$$2S(q) = \frac{q_0}{D_0} + \sum_{1 \leq i < k} \frac{1}{\zeta_i} \log \frac{D_{i-1}}{D_i} + \log D_{k-1}.$$

At $b = b_*$ this is exactly $2C(b_*)$, since $b_i = 1/D_i$ and $2h_0/b_0 = q_0/D_0$. The remaining normalization is

$$2 \left(h_{k-1} - \sum_i w_i h_i q_i \right) = \frac{q_0}{D_0} + \sum_{i=1}^{k-1} \left(\frac{q_i}{D_i} - \frac{q_{i-1}}{D_{i-1}} \right) = \frac{q_{k-1}}{D_{k-1}} = b_* - 1.$$

To check the first equality, expand the expression in parentheses in successive increments of h . The coefficient of h_0 is $1 - \sum_i w_i q_i = D_0$; the coefficient of $h_i - h_{i-1}$ is $1 - \sum_{j \geq i} w_j q_j = D_{i-1} + \zeta_i q_{i-1}$. The stationary increment formulas above then give the displayed telescoping sum. Consequently, for every stationary finite pair,

$$\lim_n l_n(h) = C(b_*) + \frac{b_* - 1}{2} - h_{k-1} = S(q) - \sum_i w_i h_i q_i.$$

Applying this to a minimizing pair proves pointwise (6).

Uniformity. With the interval weights fixed, the minimum in (6) is Lipschitz in h for the norm $\sum_i w_i |h_i|$, since $0 \leq q_i \leq 1$. The same Lipschitz bound for l_n therefore upgrades pointwise convergence to uniform convergence on compact sets.

For the second assertion, fix $B < 1$ and let $\mathcal{Q}_B$ be the nondecreasing paths taking values in $[0, B]$, modulo a.e. equality. For $q, \tilde{q} \in \mathcal{Q}_B$, the positive-part formula for D gives

$$\|D_q - D_{\tilde{q}}\|_\infty \leq \|q - \tilde{q}\|_1, \quad D_q(t) \geq 1 - B \quad (0 \leq t \leq B),$$

while $D_q(t) = 1 - t$ for $t \geq B$. Thus $S(q)$ and $q \mapsto h_q := A_q(q(\cdot))/2$ are continuous in L^1 ; the fields satisfy $0 \leq h_q \leq B/[2(1-B)^2]$. The class $\mathcal{Q}_B$ is compact in L^1 by monotone selection and bounded convergence, and its finite step paths are dense. Choose a finite net of such step paths so that both h_q in L^1 and $S(q) - \langle h_q, q \rangle$ are uniformly approximated. The common Lipschitz bound for l_n bounds the error at any q by these two approximation errors and the error at a net point. The latter errors tend to zero at the finitely many fixed stationary step pairs, as just proved. Sending $n \rightarrow \infty$ and then refining the net proves (8). In particular, no bound uniform in cascade depth is needed for the canonical log-total fluctuations. $\square$

5 The upper bound

We compare the negative expected pressure with an affine function of the pattern density and the cascade field, adapting Mourrat's supersolution argument [12, Section 4, proof of Theorem 4.1]. The cascade enrichment follows [14, Equation (1.6)]. The time derivative is now an increment from one fresh pattern, whose tail-order property supplies the comparison used below.

5.1 Enrichment, concentration, and derivatives

Fix a step trial q with maximum $q_{k-1} < 1$, and use its intervals for a cascade. For $0 \leq t \leq \alpha$, replace the fixed pattern count by an independent $K \sim \text{Pois}(Nt)$, keeping the patterns themselves standard

Gaussian. Integrate also over labels with cascade weights, and add the independent Gaussian log factor $z(\gamma) \cdot x - Nh_{k-1}$ from $l_N(h)$, where h is a nonnegative nondecreasing step field on the same intervals. The label overlap $T = (\gamma \wedge \eta)/k$ is positive semidefinite and has constant diagonal. Enumerate monomials $\kappa_j(R, T) = R^p T^d$ with p, d nonnegative integers, $p + d \geq 1$, omitting a variable at exponent zero. Let $Y_j(x, \gamma)$ be independent centered Gaussian fields with these covariances (on the index pairs, independent also of the other energies and weights). They have a hierarchical mark realization. Use independent Gaussian spin polynomial fields of covariance R^p at the vertices (Gaussian tensors on normalized spin coordinates, fields constant in spin if $p = 0$), each scaled deterministically by $\sqrt{b_i - b_{i-1}}$ at depth i , $0 \leq i \leq k - 1$, where $b_{-1} = 0$ and the b_i are the ordered covariance levels of the T^d factor (all 1 if $d = 0$). Add these increments along each path; this is done independently for the different perturbations. Thus after their use the cascade label probabilities in a two-replica average are still w_i .

Write $e_N = N^{-1/16}$. Add to the exponential's log factor the perturbation

$$\sqrt{N} e_N \sum_{j \leq N} 2^{-j} u_j Y_j, \quad 1 \leq u_j \leq 2.$$

Let $P_N(t, h, u)$ be the *random*, positively signed normalized log integral of this model and $F_N = -EP_N$. A zero perturbation can be indicated by setting $u = 0$. Uniformly, the difference of expectations upon including the perturbation is $O(e_N^2)$ by bounded covariance interpolation. For t, h on compact sets in the stated domain and the perturbation parameters in their bounds, we have

$$E|P_N - EP_N|^2 \leq C/N \tag{9}$$

with k, w fixed. To prove this, include the patterns, K , and all common Gaussian increments in the root data. Let X_0 be the value of the unnormalized backward recursion, and let $\mathcal{T}, \tilde{\mathcal{T}}$ be the original and transformed unnormalized cascade totals. The marked transformation in Section 2 gives

$$NP_N = X_0 + \log \tilde{\mathcal{T}} - \log \mathcal{T}.$$

Conditional on the root data, both totals have the ordinary fixed-cascade law. Their log difference therefore has mean zero and bounded second moment, uniformly in N, t, h, u ; independence of the two totals is not needed.

It remains to bound the variance of X_0 . Replacing one pattern changes it by at most $2\|f\|_\infty$, and adding one changes it by at most $\|f\|_\infty$. At fixed patterns, its Lipschitz constant in the common standard Gaussian coordinates is at most $C\sqrt{N}$: the energy coefficient vectors have this norm bound uniformly on states, and log integration and the backward recursion preserve it. Conditional bounded differences and Gaussian Poincaré thus give variance $O(N + K)$ at fixed count, whose expectation is $O(N)$. The conditional mean is Lipschitz in the count with the stated increment bound, so the Poisson variance Nt adds another $O(N)$. This proves (9). The fixed-size differentiations below in the u variables have the ordinary Gibbs first and second derivatives.

In the h variables, the right directional derivative of F_N when adding a nonnegative nondecreasing step direction d equals

$$E\langle R_{12} d_{\gamma^1 \wedge \gamma^2} \rangle, \tag{10}$$

by adding the corresponding covariance and using Gaussian differentiation; the bounded Gibbs expectations are continuous along such finite-parameter segments including at the starting point (by, e.g., the integral moment bounds at fixed size). In $t > 0$,

$$\partial_t F_N = -E \log \left\langle \exp f(\hat{g} \cdot x / \sqrt{N}) \right\rangle, \quad (11)$$

including expectation over the independent new pattern $\hat{g}$. This is Poisson differentiation (the log change per pattern is bounded); one can use the left derivative at the right endpoint. These expectations F_N are continuous on the parameter sets, by Poisson and Gaussian comparison or the same integral bounds.

5.2 A contact minimum forces the joint GG identities

The selection of perturbations at a contact minimum follows [12, proof of Theorem 4.1, Steps 1–3]. The curvature bound at that minimum, together with concentration, enforces the identities at the selected parameters rather than only after parameter averaging.

Suppose along a subsequence that the expected pressure exceeds $S(q) + \alpha V(q)$ by a fixed positive gap. Poissonizing at α changes it by $o(1)$: couple the counts, whose expected absolute difference is $O(\sqrt{N} + 1)$, and use the bounded log increment per pattern. At $h = 0$ and zero perturbation the label integration changes nothing. Choose $\delta > 0$ with $\alpha\delta$ smaller than the gap, and define

$$\begin{aligned} J_N(t, h, u) := & F_N(t, h, u) - \sum_i w_i q_i h_i + S(q) + t(V(q) + \delta) \\ & + \sum_{j \leq N} 2^{-j} (u_j - 3/2)^2. \end{aligned} \quad (12)$$

We minimize J_N on

$$0 \leq t \leq \alpha, \quad 0 \leq h_0 \leq \dots \leq h_{k-1} \leq H, \quad 1 \leq u_j \leq 2 \quad (j \leq N),$$

with H fixed below. Its minimum is negative and bounded away from zero for large N in the subsequence, by evaluation at $(\alpha, 0, (3/2)_j)$. At $t = 0$, the spherical dual formula and perturbation comparison give $J_N \geq -o(1)$ uniformly for fixed H .

To exclude the field cap, fix $\eta > 0$ with $q_{k-1} + \eta < 1$. Applying (6) with trial $q + \eta$ and comparing with the pattern-free model gives, uniformly on this compact region,

$$F_N(t, h, u) \geq \sum_i w_i (q_i + \eta) h_i - S(q + \eta) - \alpha \|f\|_\infty - o(1).$$

Since $|V(q)| \leq \|f\|_\infty$, it follows that

$$J_N(t, h, u) \geq \eta \sum_i w_i h_i - (S(q + \eta) - S(q)) - 2\alpha \|f\|_\infty - o(1).$$

On the cap, the sum is at least $w_{k-1}H$. Choose H so that $\eta w_{k-1}H > S(q + \eta) - S(q) + 2\alpha \|f\|_\infty + 1$, and then take N large. All cap values are positive, so any deterministic minimizing tuple (t_N, h_N, u_N) has $t_N > 0$ and $h_{N,k-1} < H$. Comparing its perturbation coefficients with their common center also bounds the quadratic penalty by $O(e_N^2)$. Thus $u_{N,j} \rightarrow 3/2$ for every fixed j .

The selected point has two first-order properties. Every nonnegative nondecreasing field direction d is feasible for a sufficiently small positive increment, and a left time increment is feasible because $t_N > 0$. Therefore

$$\begin{aligned} \mathbb{E} \langle R_{12} d_{\gamma^1 \wedge \gamma^2} \rangle & \geq \sum_i w_i q_i d_i, \\ \partial_t^- F_N(t_N, h_N, u_N) + V(q) + \delta & \leq 0. \end{aligned} \quad (13)$$

Unless a parameter is explicitly varied, all Gibbs averages in the rest of this section use these selected values. We next identify their joint overlap limits; this will contradict the time inequality in (13).

Fix j , hold the other parameters at their selected values, and write $g_N(v)$ for $\mathbb{E}P_N$ as a function of $u_j = v$, with $v_N = u_{N,j}$. The Gibbs differentiation formulas give

$$\partial_{u_j} P_N = \frac{e_N 2^{-j}}{\sqrt{N}} \langle Y_j \rangle, \quad g_N''(v_N) = e_N^2 4^{-j} E \langle (Y_j - \langle Y_j \rangle)^2 \rangle.$$

For large N , v_N is interior. Minimality gives $g_N'(v_N) = 2^{1-j}(v_N - 3/2)$. Together with convexity of g_N , the same minimality gives the finite remainder bound

$$0 \leq g_N(v_N + r) - g_N(v_N) - r g_N'(v_N) \leq 2^{-j} r^2 \quad (14)$$

whenever $v_N + r \in [1, 2]$. In particular $g_N''(v_N) \leq 2^{1-j}$, so the thermal fluctuation satisfies

$$\mathbb{E} \langle |Y_j - \langle Y_j \rangle| \rangle \leq C_j e_N^{-1}.$$

To bound the disorder fluctuation, bound the random derivative by its two convex difference quotients with step s . Use (14) for their expectations and (9) at the three parameter values. This gives

$$E |\partial_{u_j} P_N - \partial_{u_j} \mathbb{E} P_N| \leq C_j s + C/(s\sqrt{N}).$$

Taking $s = N^{-1/4}$ and using the first-derivative formula above yields

$$\mathbb{E} |\langle Y_j \rangle - \mathbb{E} \langle Y_j \rangle| \leq C_j N^{1/4} e_N^{-1}.$$

The thermal and disorder bounds are respectively $O_j(N^{1/16})$ and $O_j(N^{5/16})$, both smaller than $\sqrt{N} e_N = N^{7/16}$. Hence

$$E \langle |Y_j - E \langle Y_j \rangle| \rangle = o(\sqrt{N} e_N).$$

This self-averaging makes Gaussian integration by parts give the GG identity up to $o(1)$ for κ_j . Explicitly, for a bounded continuous joint overlap test G on $n \geq 2$ replicas, integration by parts on replica 1 yields

$$E \langle Y_j(1) G \rangle = \sqrt{N} e_N 2^{-j} u_j E \left\langle G \left(\sum_{\ell=1}^n \kappa_j(1, \ell) - n \kappa_j(1, n+1) \right) \right\rangle.$$

Use the analogous single-replica formula for $E \langle Y_j \rangle$ and the self-averaging estimate to replace the left side by $E \langle Y_j \rangle E \langle G \rangle$ at error $o(\sqrt{N} e_N)$; the constant diagonal cancels. Fixed monomials and constants span a dense class on the joint bounded domain. Thus every joint array limit of (R, T) satisfies GG with the pair as its entry and conditioning on the joint entries. Continuous tests suffice to identify these conditional distributions. Such limits exist along further subsequences by compactness in law and retain joint weak exchangeability, Gram properties for each component and their sum, and constant diagonals.

5.3 Monotone coupling and the contradiction

We use the monotone-coupling formulation of synchronization in [12, Proposition 5.2 and Theorem 5.3], based on Panchenko's synchronization argument [19, Section 4, Lemma 2].

In any such limit, R is nonnegative and ultrametric, and $R + T$ is also ultrametric, by the GG and Gram facts above; T is itself ultrametric. The two-sample law of (R, T) is then monotonically coupled. Otherwise two positive-law sets with strictly crossed orders in the two variables can be chosen from the support. The joint GG law gives positive probability of seeing such a crossing on edges 12, 13, say $R_{12} < R_{13}$ and $T_{12} > T_{13}$. Then $R_{23} = R_{12}$ and $T_{23} = T_{13}$, contradicting ultrametricity of the sum. The marginal probabilities for $T = i/k$ remain w_i . Consequently if p is the quantile of the off-diagonal spin overlap law, the conditional mean spin overlap at $T = i/k$ is the mean of p on the corresponding interval of length w_i . Indeed the conditional laws are supported in order (allowing common endpoints), so their quantiles concatenate to such a quantile p .

Write these conditional means as

$$r_i = \mathbb{E}[R_{12} \mid T_{12} = i/k] = \frac{1}{w_i} \int_{\zeta_i}^{\zeta_{i+1}} p(u) du,$$

and let r be the step path with these values. Pass the field inequality in (13) to this array limit and take $d_i = \mathbf{1}_{\{i \geq j\}}$. We obtain

$$\sum_{i \geq j} w_i (r_i - q_i) \geq 0 \quad (0 \leq j < k).$$

The tails of $r - q$ are therefore nonnegative at all interval boundaries, and hence everywhere by linear interpolation. For $s \in [\zeta_i, \zeta_{i+1}]$, monotonicity of p gives

$$\int_s^{\zeta_{i+1}} p(u) du \geq (\zeta_{i+1} - s)r_i.$$

All subsequent interval integrals of p and r agree. Consequently $\int_s^1 p \geq \int_s^1 r \geq \int_s^1 q$ for every s , and (5) gives $V(p) \leq V(q)$.

On the other hand, the time inequality in (13), together with Poisson differentiation, says at every selected finite-size point that

$$0 \geq \partial_t^- J_N(t_N, h_N, u_N) = -\mathbb{E} \log \left\langle e^{f(\widehat{g} \cdot x / \sqrt{N})} \right\rangle + V(q) + \delta.$$

Along the same array subsequence, the one-fresh-pattern calculation in Section 3 makes the right side converge to $-V(p) + V(q) + \delta \geq \delta$, a contradiction. We have proved the limit-superior upper bound for every fixed finite trial with top value below one. The finite-step approximation in Section 3 then gives the upper bound by the stated infimum.

6 Bulk covariances and the spherical identity

To add new spin coordinates, we must identify the covariance of the field they receive from the old patterns. Two-pattern removal makes this covariance a deterministic function of the spin overlap. A tangential integration identity then identifies that function with the entropy derivative from Section 4. For the related use of cavity identities for overlap observables to identify critical points in Gaussian vector-spin models, see [5, Sections 6–7].

6.1 Good perturbation parameters

For the lower bound we return to fixed pattern counts. We will average perturbation parameters when telescoping cavity increments, but first need to evaluate every subsequential limit outside a set of parameters of vanishing probability. Let $Y_p(x)$, $1 \leq p \leq N$, be independent centered Gaussian polynomial fields of covariance $R(x, x')^p$ on S_N , independent of the patterns. For a parameter sequence $v = (v_p)_{p \geq 1}$ with $1 \leq v_p \leq 2$ add the energy in the exponent

$$s_N \sum_{p \leq N} 2^{-p} v_p Y_p(x), \quad s_N = N^{3/8}, \quad \xi_N(r) = \sum_{p \leq N} 4^{-p} v_p^2 r^p. \quad (15)$$

Thus its covariance is $s_N^2 \xi_N(R)$. Averages E with v specified keep these parameters fixed. Eventually expected log-partition functions will be integrated over v with the product uniform probability π on $[1, 2]^{\mathbb{N}}$; writing their *unnormalized by dimension* integrated values as $\mathcal{F}_N$ (they use measure σ_N), we have $\mathcal{F}_N = NEp_N + o(N)$ by Gaussian comparison.

Call the perturbed system on S_N with M_N patterns the full bulk system, and call the system omitting its last two patterns the omitted system. Both are defined for all sufficiently large N . There are good parameter sets $\mathcal{G}_N$ of π -probability tending to 1 with the following property: for any deterministic sequence $v = v(N) \in \mathcal{G}_N$ (also along subsequences), every limiting overlap array of the system with two omitted satisfies scalar GG. We include an argument to be precise about parameter selection. For a fixed p and the omitted system write P_N° for its random normalized-by- N log partition. Its variance over disorder for parameters fixed even in a slightly enlarged coupling range is $O(1/N)$, by bounded differences for patterns and the Gaussian variance bound using the uniform norm of the additive energy coefficients. The deterministic function EP_N° in v_p is convex with derivative

$$(s_N 2^{-p})^2 v_p (1 - E\langle R_{12}^p \rangle) / N.$$

Its second derivative is $(s_N 2^{-p})^2 E\langle (Y_p - \langle Y_p \rangle)^2 \rangle / N$. These formulas imply, integrating over $v_p \in [1, 2]$, a uniform (in N and the other allowed parameters) bound on the integrated expectation of this thermal variance of Y_p . For the first derivative fluctuations, integrated convex difference quotients give

$$\int_1^2 E |\partial_{v_p} P_N^\circ - \partial_{v_p} EP_N^\circ| dv_p \leq C_p d s_N^2 / N + C / (d\sqrt{N})$$

for small $d > 0$. Indeed use convex upper and lower bounds on the random derivative by respectively the forward and backward difference quotients with step d , bound deviations of the three function values by the variance bound, and integrate differences of the expected derivative between points distance d apart using its monotonicity and $O_p(s_N^2/N)$ magnitude (here the constant may depend on the fixed degree p). Taking $d = N^{-1/8}$ and using $\partial_{v_p} P_N^\circ = (s_N 2^{-p}/N) \langle Y_p \rangle$ proves that

$$\int E \langle |Y_p - E\langle Y_p \rangle| \rangle \pi(dv) / s_N \longrightarrow 0.$$

Here is an explicit diagonal selection. Denote these normalized nonnegative deviations by $d_{N,p}(v)$. Choose $J_N \uparrow \infty$ slowly enough that $J_N \leq N$ and $\sum_{p \leq J_N} \int d_{N,p} d\pi \rightarrow 0$; call the sum a_N . With $\eta_N = \sqrt{a_N} + N^{-1}$, set

$$\mathcal{G}_N = \{v : \max_{p \leq J_N} d_{N,p}(v) \leq \eta_N\}.$$

Markov's inequality gives $\pi(\mathcal{G}_N^\varepsilon) \leq a_N/\eta_N \rightarrow 0$, and each fixed-degree deviation tends to zero uniformly on $\mathcal{G}_N$. Integration by parts as for the joint perturbation, now testing with all powers R^p , proves the property. We only require it for the omitted system.

6.2 The bulk limit and two-pattern removal

For $x \in S_N$ let $b_a(x) = g^a \cdot x/\sqrt{N}$. In the full perturbed bulk define the sample observables

$$\begin{aligned} B_N(x, x') &= \frac{1}{N} \sum_{a=1}^{M_N} f'(b_a(x)) f'(b_a(x')), \\ C_N(x) &= \frac{1}{N} \sum_{a=1}^{M_N} (f''(b_a(x)) - b_a(x) f'(b_a(x))). \end{aligned} \tag{16}$$

These are uniformly bounded because f is smooth and compactly supported. The next proposition supplies all the bulk data needed in the cavity calculation.

Proposition 5 (Identification of the bulk limit). *Choose any deterministic parameters $v(N) \in \mathcal{G}_N$, with N running along any subsequence tending to infinity. Suppose that the omitted system's overlap array and the full system's arrays*

$$(R_{\ell j}, B_N(x^\ell, x^j), C_N(x^\ell))_{\ell, j \geq 1}$$

converge in law in all finite restrictions, under their respective disorder and Gibbs averages. Let μ be the off-diagonal two-replica law of the omitted limiting array, and let q be its quantile. Then the limiting full and omitted overlap arrays have the same law. There are constants a_d, c and a deterministic nonnegative nondecreasing function a , specified μ -almost everywhere, such that the full-system limit satisfies almost surely

$$\begin{aligned} B_{\ell j} &= a(R_{\ell j}) \quad (\ell \neq j), \quad B_{\ell \ell} = a_d, \quad C_\ell = c, \\ 0 &\leq a \leq a_d \leq \alpha \|f'\|_\infty^2. \end{aligned} \tag{17}$$

Moreover,

$$\begin{aligned} c + a_d &= \int r a(r) \mu(dr), \\ a(r) &= A_q(r) = \int_0^r \frac{dt}{D_q(t)^2} \quad (\mu\text{-a.e. } r), \\ \sup \sup \mu &\leq B_* := \frac{\alpha \|f'\|_\infty^2}{1 + \alpha \|f'\|_\infty^2} < 1. \end{aligned} \tag{18}$$

The law μ and the quantities a, a_d, c may depend on the chosen subsequential limit; the bound B_* does not.

We prove the covariance identification first, and then derive the spherical identities in the next subsection. Fix a sequence and limits as in the proposition. The omitted limiting overlap array satisfies GG by the construction of $\mathcal{G}_N$; in particular its off-diagonal entries are nonnegative and ultrametric. Write E_{lim} for expectation in the full-system limiting array law.

Restoring the two omitted patterns reweights the omitted Gibbs measure by $e^{f(b_1^*) + f(b_2^*)}$, where the stars denote independent fresh Gaussian marks. This factor is bounded above and away from zero. The fresh-field and reciprocal-polynomial calculation of Section 2 therefore computes every bounded continuous finite-replica limit involving the restored marks from the omitted overlap

array. Round that limiting array's off-diagonal values down to finitely many levels, with vanishing supremum error, retaining diagonal one. Each rounded array has a finite RPC representation, and its marked calculations converge to the unrounded ones.

The marked transformation preserves the rounded RPC overlap law. It also keeps the two restored mark components independent conditional on the entire selected finite label shape. Indeed their leaf log factors add, their backward recursions add, and their tilted child kernels factor. The transformed unmarked tree is independent of the root marks, whose original product law is retained, as proved in Section 2. The residual Gaussian at a leaf is sampled afresh on every replica visit, including repeated visits to that leaf. Thus the limiting full and omitted overlap arrays have the same law after passing through the roundings.

For one restored mark, let M_i be the conditional mean of its terminal observable f' given the path through level i , under the tilted single-path law. Use distinct overlap values of positive probability as the levels. Given a replica pair whose common level is i , its two below-fork paths are independent conditional on the common path; at the leaf level the residual draws are still independent. The product of the two terminal f' values has conditional mean M_i^2 given the common path. Averaging over that path gives the conditional mean given the common level i , namely the deterministic number

$$d_i = E[M_i^2].$$

The conditional means M_i form a bounded martingale, so the d_i are nonnegative nondecreasing and bounded by the one-path second moment $d_d = E[f'(b^*)^2] \leq \|f'\|_\infty^2$. These conclusions also hold conditional on the whole sampled label shape, since the tilted kernels do not depend on labeling indices. For the two restored marks, component independence therefore gives d_i^2 as the conditional mean of the product of their pair observables. Likewise, each one-path observable f'^2 or $f''(b) - bf'(b)$ has a constant mean given labels, and products using the two restored marks have factorized means.

For each rounding, extend the values d_i monotonically across its bins, with the same bounds in empty bins. Along a rounding subsequence these functions converge μ -almost everywhere to a bounded nondecreasing function d : use monotone selection and a further diagonal extraction at the atoms of μ . Extract also the bounded one-path means, and put $a = \alpha d$. Pattern exchangeability then gives, for every continuous $\psi : [-1, 1] \rightarrow \mathbb{R}$,

$$\begin{aligned} E_{\text{lim}}[B_{12}\psi(R_{12})] &= \int a(r)\psi(r) \mu(dr), \\ E_{\text{lim}}[B_{12}^2] &= \int a(r)^2 \mu(dr). \end{aligned} \tag{19}$$

To see the multipliers explicitly, the finite- N first moment uses one designated pattern with coefficient M_N/N , and the second uses two distinct designated patterns with coefficient $M_N(M_N - 1)/N^2$. Both designated patterns can be the last two; the repeated-pattern terms in B_N^2 are $O(N^{-1})$. First pass to the bulk limit in these restored-pattern expectations, then evaluate them by the rounded cascades and use bounded convergence along the rounding subsequence. This proves (19) without assuming that a is continuous.

The first identity in (19) identifies $E_{\text{lim}}[B_{12} | R_{12}] = a(R_{12})$, since continuous tests determine the corresponding finite signed measure. The second therefore yields

$$E_{\text{lim}}[(B_{12} - a(R_{12}))^2] = 0.$$

The same first- and second-moment argument for the one-path observables gives constants $B_{\ell\ell} = a_d$ and $C_\ell = c$ with zero variance. The bounds in (17) follow from those for d_i, d_d . Replica exchangeability and countability give the asserted identities simultaneously for the whole limiting array.

6.3 Spherical self-consistency

It remains to prove (18). These identities will identify the covariance just obtained with the stationary field in the spherical formula.

The first equality follows from Gaussian integration by parts on a pattern in the full finite- N Gibbs average: the average of $b_a(x^1)f'(b_a(x^1))$ under $E\langle \cdot \rangle$ equals the average of

$$f''(b_a(x^1)) + f'(b_a(x^1))^2 - R_{12}f'(b_a(x^1))f'(b_a(x^2)).$$

Sum, divide by N and pass to (16)–(17).

For a smooth scalar test $G(R_{12})$, use integration of tangential divergence on the sphere in the variable x^1 , with tangent direction $u = x^2 - R_{12}x^1$. Its divergence on S_N is $-(N-1)R_{12}$ (tangent projection of a constant vector on a sphere of radius $\sqrt{N}$). If H denotes the full log density up to a constant, this gives

$$(N-1)E\langle R_{12}G \rangle = E\langle G'(R_{12})(1 - R_{12}^2) + G u \cdot \nabla H(x^1) \rangle.$$

This identity is first at fixed disorder with Gibbs integration in the sphere variables. The perturbation part of the last gradient term contributes $O_G(s_N^2)$ in expectation. In fact after Gaussian integration by parts its expectation with the test uses derivatives of the covariance $s_N^2 \xi_N(R_{1j})$ in the first variable in direction u : it is

$$s_N^2 E \left\langle G \left(\sum_{j=1}^2 \xi'_N(R_{1j})(R_{2j} - R_{12}R_{1j}) - 2\xi'_N(R_{13})(R_{23} - R_{12}R_{13}) \right) \right\rangle.$$

All derivatives of ξ_N used here are uniformly bounded on $[-1, 1]$. The contribution of pattern a uses $f'(b_a(x^1))(b_a(x^2) - R_{12}b_a(x^1))$. Integrating by parts over this pattern with the two Gibbs replicas at finite N , the covariance of the tangential difference with $b_a(x^1)$ at fixed spins is zero. Only the second replica weight and the denominator terms survive; the resulting average for this pattern contribution (including G) has integrand

$$Gf'(b_a(x^1)) \left(f'(b_a(x^2))(1 - R_{12}^2) - 2f'(b_a(x^3))(R_{23} - R_{12}R_{13}) \right).$$

These calculations hold for the finite smooth densities and polynomial perturbations, with differentiations under Gaussian integration justified at each fixed size by compact spins, smooth bounded pattern terms and Gaussian moments, or the bounds in Section 2.4. Hence dividing by N , taking the limit and using (17), we find the following equality conditional on $R_{12} = r$, for μ -almost every r :

$$r = a(r)(1 - r^2) - 2E_{\lim}[a(R_{13})(R_{23} - rR_{13}) \mid R_{12} = r]. \quad (20)$$

Equality against all smooth tests gives this conditional identity under the limiting array law. The test-derivative term is $O_G(N^{-1})$ after division by N , and the perturbation term is $O_G(s_N^2/N) = o(1)$.

By GG, R_{13} and R_{23} each conditionally have law $(\mu + \delta_r)/2$. If $R_{13} < r$, ultrametricity forces $R_{23} = R_{13}$; if $R_{13} > r$, $R_{23} = r$; also $R_{13} = r$ whenever $R_{23} > r$. Consequently

$$\begin{aligned} 2E_{\lim}[a(R_{13})R_{23} \mid R_{12} = r] &= r \left(\int a \, d\mu + a(r) \right) - \int (r - s)_+ a(s) \mu(ds) + a(r) \int (s - r)_+ \mu(ds), \\ 2E_{\lim}[a(R_{13})R_{13} \mid R_{12} = r] &= \int sa(s) \mu(ds) + ra(r). \end{aligned}$$

Combining with (20),

$$a(r)D_q(r) = U(r) \quad (\mu\text{-a.e.}), \quad U(r) = r \left(1 + \int (1-s)a(s)\mu(ds) \right) - \int (r-s)_+ a(s)\mu(ds). \quad (21)$$

Taking $D_q(1) = 0$, we have $U(1) = 1$. Since U, D_q are continuous and a is bounded, there is $\varepsilon > 0$ such that $U(r) > \|a\|_\infty D_q(r)$ on $[1-\varepsilon, 1]$, where the norm is with respect to μ . Equation (21) therefore implies $\mu([1-\varepsilon, 1]) = 0$. In particular, the support of μ lies strictly below 1, and $D_q(r) > 0$ for every $r < 1$.

We can now solve (21) on the whole interval $[0, 1)$. The continuous Lipschitz functions D_q, U have distributional second derivatives $D_q'' = -\mu$ and $U'' = -a\mu$ on $(0, 1)$. The product rule for their bounded-variation derivatives gives, as an identity of measures,

$$d(U'D_q - UD_q') = (U - aD_q)\mu = 0.$$

This includes atoms: at an interior atom r , the jump is $[U(r) - a(r)D_q(r)]\mu\{r\} = 0$. Thus $U'D_q - UD_q'$ is constant. Above the support, $D_q(r) = 1 - r$ and U is affine with $U(1) = 1$, so this constant is 1. Consequently U/D_q is absolutely continuous on every compact subinterval of $[0, 1)$, with derivative D_q^{-2} . Since $U(0) = 0$,

$$\frac{U(r)}{D_q(r)} = \int_0^r \frac{dt}{D_q(t)^2} = A_q(r).$$

Together with (21), this proves the second identity in (18), including $r = 0$ if it is an atom of μ .

Finally, $D_q(t) \leq 1 - t$ gives $A_q(r) \geq r/(1-r)$. Combining this with $a(r) \leq \alpha\|f'\|_\infty^2$ proves the support bound in (18). The continuous increasing function A_q agrees with a μ -almost everywhere and is bounded by a_d on the entire support, by continuity. It is therefore also bounded by a_d at every nonnegative value obtained by rounding an overlap down. In the cavity calculation we may use this representative for the off-diagonal covariance: the field parameters are $2h = A_q(q)$, exactly as in (7), and their rounded values remain valid covariance levels below the diagonal variance a_d .

This completes the proof of Proposition 5.

7 The cavity lower bound

Following the fixed-block cavity approach of [1], we add a fixed number L of spin coordinates. Proposition 5 supplies the covariance data needed for the uniform spherical formula in Proposition 4. We first take the bulk dimension to infinity, average the perturbation parameters so that increments telescope, and only then send L to infinity.

7.1 A uniform increment inequality

Fix a positive integer L , and compare expected log partition functions with the perturbation (15), in dimensions $N + L$ and N with their full pattern counts and sphere probability measures, at the *same* values of the coefficients v . Denote this difference at given v by $\Delta_{N,L}(v)$. Put $d = M_{N+L} - M_N$ (possibly depending on N). In dimension $N + L$ write a spin in terms of cavity coordinates $z \in \mathbb{R}^L$ and the direction of its other coordinates, the latter parameterized by $x \in S_N$. Ignore the null event of vanishing other coordinates. Under the original surface measure, x has law σ_N independent of z , and the law of z , denoted $\nu_{N,L}$, is spherically symmetric and converges to standard Gaussian

for fixed L (by the normalized Gaussian representation). We restrict the integral to the shell $I_L = \{z : L \leq \|z\|^2 \leq L+1\}$, giving a lower bound. There $\nu_{N,L}(I_L) \rightarrow p_L := \mathbb{P}(\chi_L^2 \in [L, L+1]) > 0$.

If $g^a \in \mathbb{R}^N$ and $y_a \in \mathbb{R}^L$ denote the two independent standard Gaussian blocks of pattern a , its field on such a spin is

$$\rho_z b_a(x) + \frac{y_a \cdot z}{\sqrt{N+L}}, \quad \rho_z = \sqrt{1 - \|z\|^2/(N+L)}$$

using b_a as before also for the d new patterns. The perturbation on the restricted integral can be replaced in expected log partition by exactly the bulk perturbation (15) on x , up to $o(1)$ for fixed L , uniformly in v . Indeed the old and desired Gaussian covariances are

$$s_{N+L}^2 \xi_{N+L} \left(\rho_z \rho_{z'} R(x, x') + \frac{z \cdot z'}{N+L} \right), \quad s_N^2 \xi_N(R(x, x')),$$

whose uniform difference on the restriction goes to zero, by the power bound for the change of s_N^2 , uniform derivative bounds for ξ and the summable-coefficient tail here decaying exponentially. Interpolate conditionally on the patterns.

Conditional on the bulk data define a centered Gaussian vector field $\mathcal{Y}(x) \in \mathbb{R}^L$, with independent coordinates of covariance $B_N(x, x')$, by

$$\mathcal{Y}(x) = N^{-1/2} \sum_{a=1}^{M_N} f'(b_a(x)) y_a.$$

For fixed L , Taylor expansion on the shell for the original patterns gives their log factor sum there as

$$\begin{aligned} \sum_{a \leq M_N} f(b_a(x)) + z \cdot \mathcal{Y}(x) + \frac{\|z\|^2}{2} C_N(x) + Q(x, z) + \text{rem}(x, z), \\ Q(x, z) = \frac{1}{2N} \sum_{a=1}^{M_N} f''(b_a(x)) ((y_a \cdot z)^2 - \|z\|^2). \end{aligned} \tag{22}$$

where $E \sup_{x,z} |\text{rem}(x, z)| = o(1)$ on the restriction. To verify the uniform bound, compact support of derivatives gives uniformly in real s

$$f(\rho_z s) = f(s) - \frac{\|z\|^2}{2N} s f'(s) + O_L(N^{-2}), \quad f^{(j)}(\rho_z s) = f^{(j)}(s) + O_L(N^{-1}) \quad (j = 1, 2);$$

these use bounded derivatives after multiplication by the powers of s needed in Taylor expansion of the dilation near 1. Adding $y_a \cdot z / \sqrt{N+L}$ with the bounded third derivative remainder and adjusting denominators to N adds errors bounded by constants depending on L, f times $(\|y_a\| + \|y_a\|^3) N^{-3/2} + \|y_a\|^2 N^{-2}$ per pattern, in addition to the $O_L(N^{-2})$. Similarly the log factors of the d new patterns can be replaced by $f(b_a(x))$ with total expected uniform error $o(1)$, using $d = O_L(1)$.

In a lower bound on expected log partition after these replacements we can drop Q at cost $o(1)$ for fixed L . Indeed use $\log \langle e^Q \rangle_{\text{app}} \geq \langle Q \rangle_{\text{app}}$ in the approximating restricted Gibbs measure omitting Q . This measure is from the full N -bulk measure and the normalized shell probability by a tilt with log

$$z \cdot \mathcal{Y}(x) + \|z\|^2 C_N(x)/2 + \sum_{a=M_N+1}^{M_N+d} f(b_a(x)).$$

The denominator of this tilt is bounded below deterministically by a positive constant at fixed L , since only the linear term can be unbounded and averaging its exponential in the z -direction alone gives at least 1. The y_a are independent of the base bulk. For fixed x, z and that data, Q has conditional second moment $O_L(1/N)$, and $\exp(z \cdot \mathcal{Y}(x))$ has conditional second moment bounded in terms of L . Cauchy-Schwarz inside the base integration gives $E\langle |Q| \rangle_{\text{app}} = o(1)$.

Replacing $\|z\|^2 C_N/2$ by $LC_N/2$ costs at most a fixed constant C_0 independent even of L , by (16). For each x , the uniform-direction integral of the linear exponential is nondecreasing in radius, so we can now bound below using σ_L at radius $\sqrt{L}$. Subtracting the expected bulk log partition gives

$$\begin{aligned} \Delta_{N,L}(v) \geq & \log \nu_{N,L}(I_L) - C_0 - o_N(1) \\ & + E \log \left\langle e^{LC_N(x)/2 + \sum_{a=M_N+1}^{M_N+d} f(b_a(x))} \int_{S_L} \exp(z \cdot \mathcal{Y}(x)) d\sigma_L(z) \right\rangle_N, \end{aligned} \quad (23)$$

where $\langle \rangle_N$ is the full dimension- N perturbed bulk Gibbs measure (without the d new patterns or any y_a). Errors here tend to zero for fixed L uniformly in allowed v . In particular there is a uniform lower bound on $\Delta_{N,L}(v)$ for large N at fixed L .

7.2 Evaluation of the cavity term

We evaluate the limit of the last line along a good sequence with the array convergence above and, by extraction, a fixed value of d . Choose a constant K_C with $|C_N(x)| \leq K_C$ uniformly in N, x , and put

$$K_B = \alpha \|f'\|_\infty^2, \quad C_L = LK_C/2 + (\alpha L + 1)\|f\|_\infty, \quad F_L(y) = \int_{S_L} e^{z \cdot y} d\sigma_L(z).$$

The spherical factor F_L is unbounded but at least one. Let A be the Gibbs integral in the last line of (23), and let A_Λ replace F_L by $\min(F_L, \Lambda)$, where $\Lambda \geq 1$. The other logarithmic factors have absolute value at most C_L , so $A, A_\Lambda \geq e^{-C_L}$. Conditional Gaussianity and Jensen give $E\langle F_L(\mathcal{Y})^2 \rangle \leq e^{2LK_B}$. Consequently

$$0 \leq E(\log A - \log A_\Lambda) \leq e^{2C_L} E\langle (F_L(\mathcal{Y}) - \Lambda)_+ \rangle \leq \frac{e^{2C_L + 2LK_B}}{\Lambda}.$$

The same estimate holds for the rounded limiting Gaussian arrays, whose vector diagonal remains $a_d \leq K_B$. Thus at fixed L we may first use the bounded finite-replica calculation and then remove the cap uniformly. In that bounded calculation the cap is applied to the full vector mark before its residual noises are averaged; the explicit residual formula below is used only for the uncapped finite cascades.

Explicitly, the cavity vector and d additional scalar marks have conditionally independent field components given base data; their covariance arrays on base replicas are given by B_N and R , respectively. The limiting C_N entry is c . Use (17)–(18), and approximate by rounding off-diagonal overlaps down with vanishing sup error and replacing their a -values by the continuous A_q at rounded overlaps; retain the true limiting diagonals $a_d, 1$. These rounded joint covariances have an RPC realization because both ordered off-diagonal sequences are nonnegative nondecreasing bounded by the respective diagonals. They converge on the replica arrays to the desired covariances. For clarity, the capping and polynomial argument first identifies the finite- N log limit for capped integrands from the limiting array law, and also shows convergence to this same capped value from the rounded-array calculations. The uniform uncapping estimate gives the uncapped log limit by evaluating the uncapped finite cascades.

In such a cascade write $\bar{q}$ for the rounded quantile and $\bar{h} = A_q(\bar{q})/2$; the top shared vector mark has variance $2\bar{h}_{\text{top}}$ per coordinate. Averaging its independent residual noise at the leaf multiplies

the spherical integral by $\exp(L(a_d - 2\bar{h}_{\text{top}})/2)$ since $\|z\|^2 = L$. Average residual noises of new scalar marks independently as well. The log-factor recursions separate over independent components consisting of the vector field and each scalar mark field. Thus the resulting expected log Gibbs integral is exactly

$$L((c + a_d)/2 + l_L(\bar{h})) + dV(\bar{q}).$$

Using continuity on the rounding limit, the last line of (23) therefore converges to

$$L((c + a_d)/2 + l_L(h)) + dV(q), \quad h = A_q(q)/2. \quad (24)$$

By (18), $(c + a_d)/2 = \int_0^1 qh$. Uniformly for the possible limits here, (8) with B_* thus bounds (24) from below by

$$L(S(q) + \alpha V(q) - \epsilon_L) - \|f\|_\infty, \quad \epsilon_L \longrightarrow 0, \quad (25)$$

where we used the at-most-one rounding discrepancy in the number of patterns and boundedness of V .

7.3 Averaging and telescoping

Let $P = \inf_q(S(q) + \alpha V(q))$ for this terminal f . The compact-subsequence argument starting from arbitrary good parameter sequences and (23)–(25) shows, for fixed L ,

$$\liminf_N \inf_{v \in \mathcal{G}_N} \Delta_{N,L}(v) \geq \log p_L - C_0 + L(P - \epsilon_L) - \|f\|_\infty.$$

The bad sets have vanishing π -probability. To control their contribution, note that the spherical factor in (23) is at least one, so for every v and all sufficiently large N ,

$$\Delta_{N,L}(v) \geq \log(p_L/2) - C_0 - 1 - C_L.$$

This fixed- L lower bound controls their contribution. Only the omitted system in dimension N needs good parameters: the dimension- $N + L$ partition enters solely through the uniform restriction inequality. The same parameter sequence and product law in both dimensions give $\int \Delta_{N,L} d\pi = \mathcal{F}_{N+L} - \mathcal{F}_N$ exactly. Hence the same limit-inferior bound holds for this averaged increment. Telescoping for each residue modulo L gives

$$\liminf_{N \rightarrow \infty} \frac{\mathcal{F}_N}{N} \geq P - \epsilon_L + \frac{\log p_L - C_0 - \|f\|_\infty}{L}.$$

Here $\log p_L = o(L)$, as follows from the χ_L^2 density and Stirling's formula. Sending L to infinity and removing the negligible per-coordinate perturbation proves the lower bound for the original expected pressure.

8 Bounded continuous terminal data and concentration

The two pressure bounds have so far been proved for smooth compactly supported potentials. We now approximate both the expected pressure and the control value uniformly in their respective parameters.

Fix a bounded continuous $f = \beta\phi$, and choose $f_j \in C_c^\infty(\mathbb{R})$ converging to f uniformly on compact sets, with $|f_j|, |f| \leq K_0$ for a common constant K_0 . Write $p_N[g]$ for the normalized pressure with

terminal log reward g , so the pressure in Theorem 1 is $p_N[f]$. Interpolate between f and f_j in the exponent. The derivative of expected pressure is the sum, divided by N , of the expected Gibbs averages of $(f_j - f)(b_a(x))$. Delete pattern a from the interpolated Gibbs density. For any nonnegative observable, its Gibbs expectation after restoring this bounded log factor is at most e^{2K_0} times the expectation before restoration. Without that pattern, its Gaussian vector is independent of the sampled spin, so $b_a(x)$ has the standard normal law. Integrating the derivative bound and using $M_N/N \leq \alpha$ gives

$$\sup_N |\mathbb{E} p_N[f_j] - \mathbb{E} p_N[f]| \leq \alpha e^{2K_0} \mathbb{E} |f_j(G) - f(G)| \xrightarrow{j \rightarrow \infty} 0, \quad G \sim \mathcal{N}(0, 1).$$

The convergence follows from bounded convergence.

For the control values, the common payoff bound allows us to restrict every supremum to controls satisfying

$$\mathbb{E}_B \int_0^1 m(t) v_t^2 dt \leq K_1, \quad K_1 := 4K_0 + 2.$$

Indeed a control outside this set has payoff less than $-K_0 - 1$, whereas zero control gives at least $-K_0$. For these controls, Cauchy–Schwarz and $\int_0^1 m(t) dt \leq 1$ give the uniform second-moment bound

$$X_{m,v} := B_1 + \int_0^1 m(t) v_t dt, \quad \mathbb{E}_B |X_{m,v}|^2 \leq 2 + 2\mathbb{E}_B \int_0^1 m(t) v_t^2 dt \leq 2 + 2K_1.$$

Comparing the two payoffs on this common set of controls, splitting at $|X_{m,v}| = R$, and using the second-moment bound yields, for every $R > 0$,

$$\sup_{m \in \mathcal{U}} |V_{f_j}(m) - V_f(m)| \leq \sup_{|x| \leq R} |f_j(x) - f(x)| + \frac{4K_0(1 + K_1)}{R^2}.$$

First send j to infinity and then R to infinity. The resulting uniform convergence of the values implies convergence of the variational infima with the common entropy term $S(m)$. Together with the uniform pressure estimate and the smooth case, this proves the expected-pressure identity for f .

Finally, replacing any one independent pattern changes Np_N by at most $2\|f\|_\infty$. The bounded-difference variance estimate therefore gives $\text{Var}(p_N) = O(1/N)$. Combining this concentration with the expected-pressure limit proves convergence in probability and completes Theorem 1.

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